

Task 9: Session 9 (AIML) - Assignment Questions

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Domain: AIML

College & Branch Name: Zeal Polytechnic, Diploma In AIML

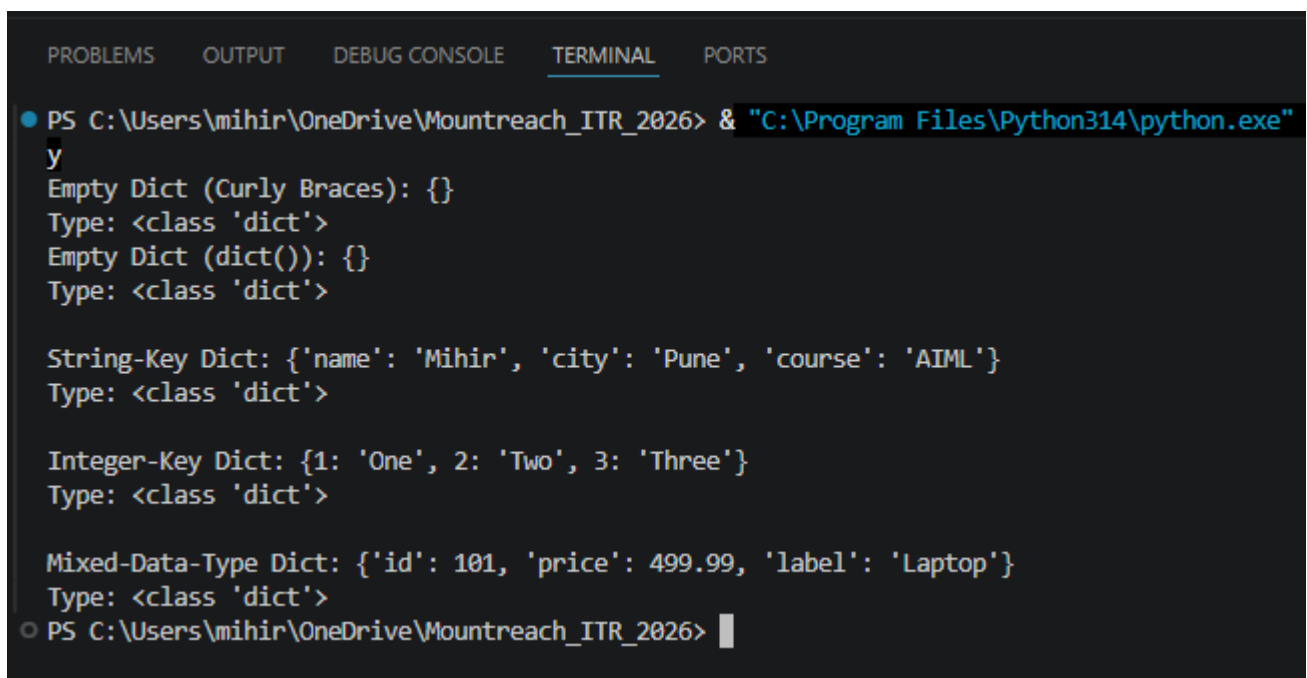
1. Creating Dictionaries

Write a program to create dictionaries using different methods:

- An empty dictionary using two different ways.
- A dictionary with string keys (name, city, course).
- A dictionary with integer keys.
- A mixed data type dictionary (string, int, float).

Print each dictionary and its type. Add comments explaining the syntax.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
y
Empty Dict (Curly Braces): {}
Type: <class 'dict'>
Empty Dict (dict()): {}
Type: <class 'dict'>

String-Key Dict: {'name': 'Mihir', 'city': 'Pune', 'course': 'AIML'}
Type: <class 'dict'>

Integer-Key Dict: {1: 'One', 2: 'Two', 3: 'Three'}
Type: <class 'dict'>

Mixed-Data-Type Dict: {'id': 101, 'price': 499.99, 'label': 'Laptop'}
Type: <class 'dict'>
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

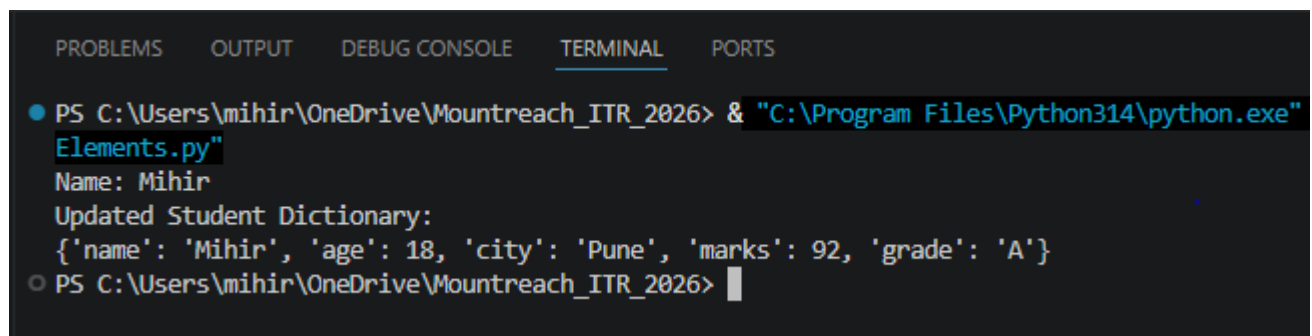
2. Accessing & Modifying Elements

Create the dictionary:

```
student = {  
    "name": "YourName",  
    "age": 00,  
    "city": "YourCity",  
    "marks": 00  
}
```

- Print the value of "name" using square brackets.
- Update the "marks" to 92.
- Add a new key "grade" with value "A".
- Print the updated dictionary.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"  
  Elements.py"  
  Name: Mihir  
  Updated Student Dictionary:  
  {'name': 'Mihir', 'age': 18, 'city': 'Pune', 'marks': 92, 'grade': 'A'}  
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

3. get(), keys(), values(), items()

Create a dictionary:

```
person = {"name": "Priya", "age": 21, "profession": "Engineer"}
```

Demonstrate:

- Using get() to access "age" and a non-existing key "salary" (with default value).
- Print all keys using keys().
- Print all values using values().
- Print all key-value pairs using items().

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
  items().py
  Age (Via get()): 21
  Salary (Via get(), Default Used): Not Available
  Keys: dict_keys(['name', 'age', 'profession'])
  Values: dict_values(['Priya', 21, 'Engineer'])
  Items: dict_items([('name', 'Priya'), ('age', 21), ('profession', 'Engineer')])
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026>
```

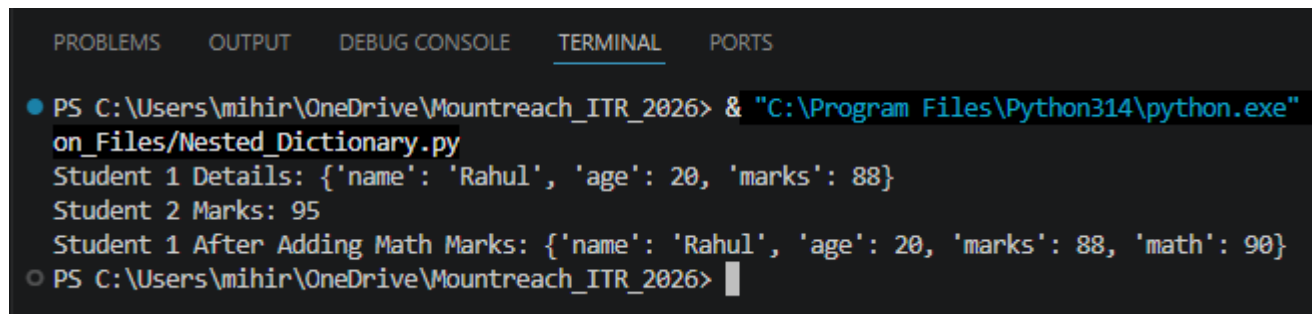
4. Nested Dictionary

Create a nested dictionary to store information of two students:

```
students = {  
    "s1": {"name": "Rahul", "age": 20, "marks": 88},  
    "s2": {"name": "Sneha", "age": 21, "marks": 95}  
}
```

- Print the details of the first student.
- Print the marks of the second student.
- Add a new subject "math" with marks 90 for the first student.

Output:



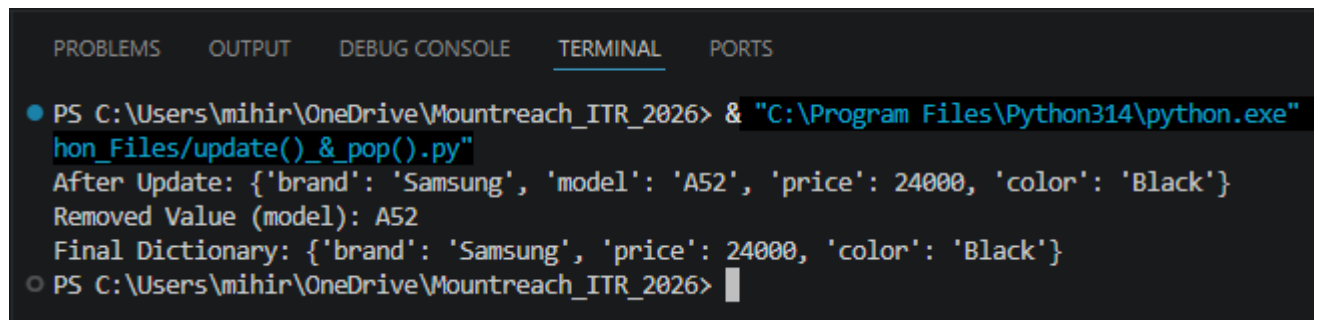
```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"  
on_Files/Nested_Dictionary.py  
Student 1 Details: {'name': 'Rahul', 'age': 20, 'marks': 88}  
Student 2 Marks: 95  
Student 1 After Adding Math Marks: {'name': 'Rahul', 'age': 20, 'marks': 88, 'math': 90}  
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> █
```

5. update() & pop()

Create a dictionary: info = {"brand": "Samsung", "model": "A52", "price": 25000}

- Update it with new information: {"color": "Black", "price": 24000}.
- Remove the key "model" using pop() and print the removed value.
- Print the final dictionary.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
  hon_Files/update()_&_pop().py"
  After Update: {'brand': 'Samsung', 'model': 'A52', 'price': 24000, 'color': 'Black'}
  Removed Value (model): A52
  Final Dictionary: {'brand': 'Samsung', 'price': 24000, 'color': 'Black'}
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> 
```

6. popitem() & clear()

Create a dictionary with at least 5 key-value pairs.

- Use popitem() twice and print the removed items each time.
- Clear the entire dictionary using clear().
- Print the final result.

Add comments explaining the difference between pop() and popitem().

Output:

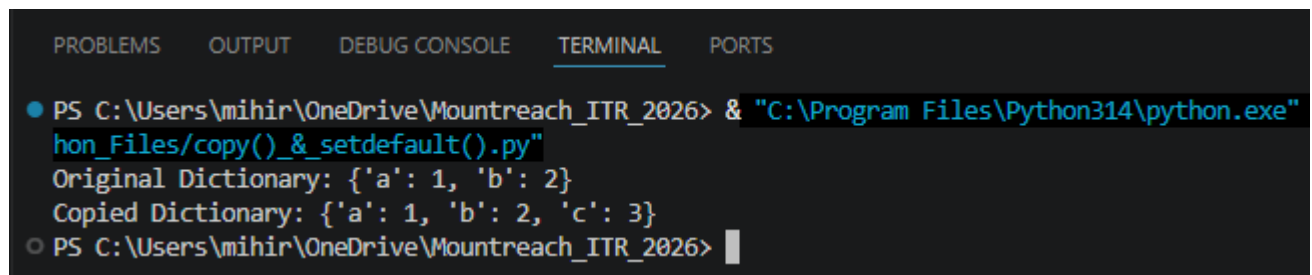
```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
  hon_Files/popitem()_&_clear().py"
Original Dictionary: {'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5}
First popitem() Removed: ('e', 5)
Second popitem() Removed: ('d', 4)
Dictionary After Two popitem() Calls: {'a': 1, 'b': 2, 'c': 3}
Dictionary After clear(): {}
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

7. copy() & setdefault()

Create a dictionary: `d = {"a": 1, "b": 2}`

- Make a copy of it.
- Use `setdefault()` to add key "c" with value 3 (if it doesn't exist).
- Use `setdefault()` again for an existing key "a".
- Print the original and copied dictionaries.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
  hon_Files/copy()_&_setdefault().py"
  Original Dictionary: {'a': 1, 'b': 2}
  Copied Dictionary: {'a': 1, 'b': 2, 'c': 3}
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```

8. fromkeys() & Membership

Write a program that:

- Uses dict.fromkeys() to create a dictionary with keys ["name", "age", "city"] and default value None.
- Takes user input to fill these keys.
- Checks if a given key exists using in operator.

Output:

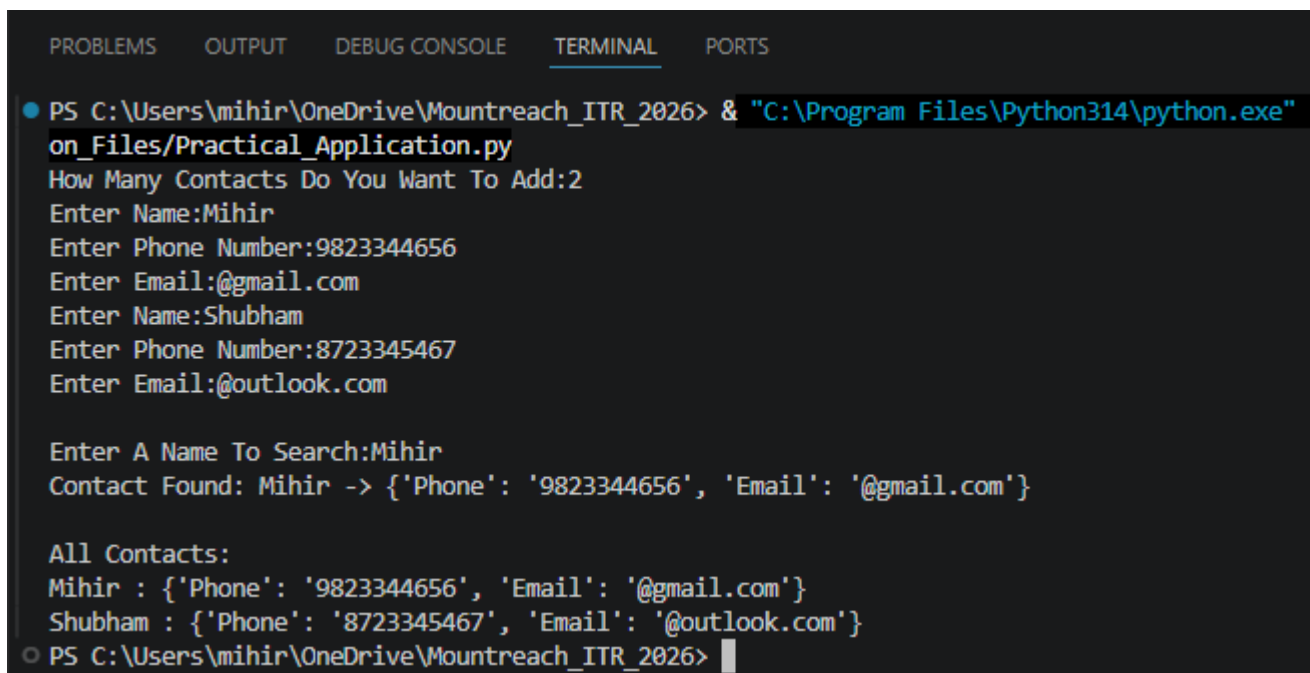
```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
  hon_Files/fromkeys() & membership.py"
Initial Dictionary: {'name': None, 'age': None, 'city': None}
Enter Your name:Mihir
Enter Your age:18
Enter Your city:Pune
Filled Dictionary: {'name': 'Mihir', 'age': '18', 'city': 'Pune'}
Enter A Key To Check If It Exists: age
age Exists In The Dictionary.
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> |
```


9. Practical Application

Write a program to store and manage contact information using a dictionary:

- Take name, phone number, and email as input.
- Store them in a dictionary with name as key.
- Allow the user to search a contact by name using get().
- Print all contacts using items().

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
on_Files/Practical_Application.py
How Many Contacts Do You Want To Add:2
Enter Name:Mihir
Enter Phone Number:9823344656
Enter Email:@gmail.com
Enter Name:Shubham
Enter Phone Number:8723345467
Enter Email:@outlook.com

Enter A Name To Search:Mihir
Contact Found: Mihir -> {'Phone': '9823344656', 'Email': '@gmail.com'}

All Contacts:
Mihir : {'Phone': '9823344656', 'Email': '@gmail.com'}
Shubham : {'Phone': '8723345467', 'Email': '@outlook.com'}
○ PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> █
```

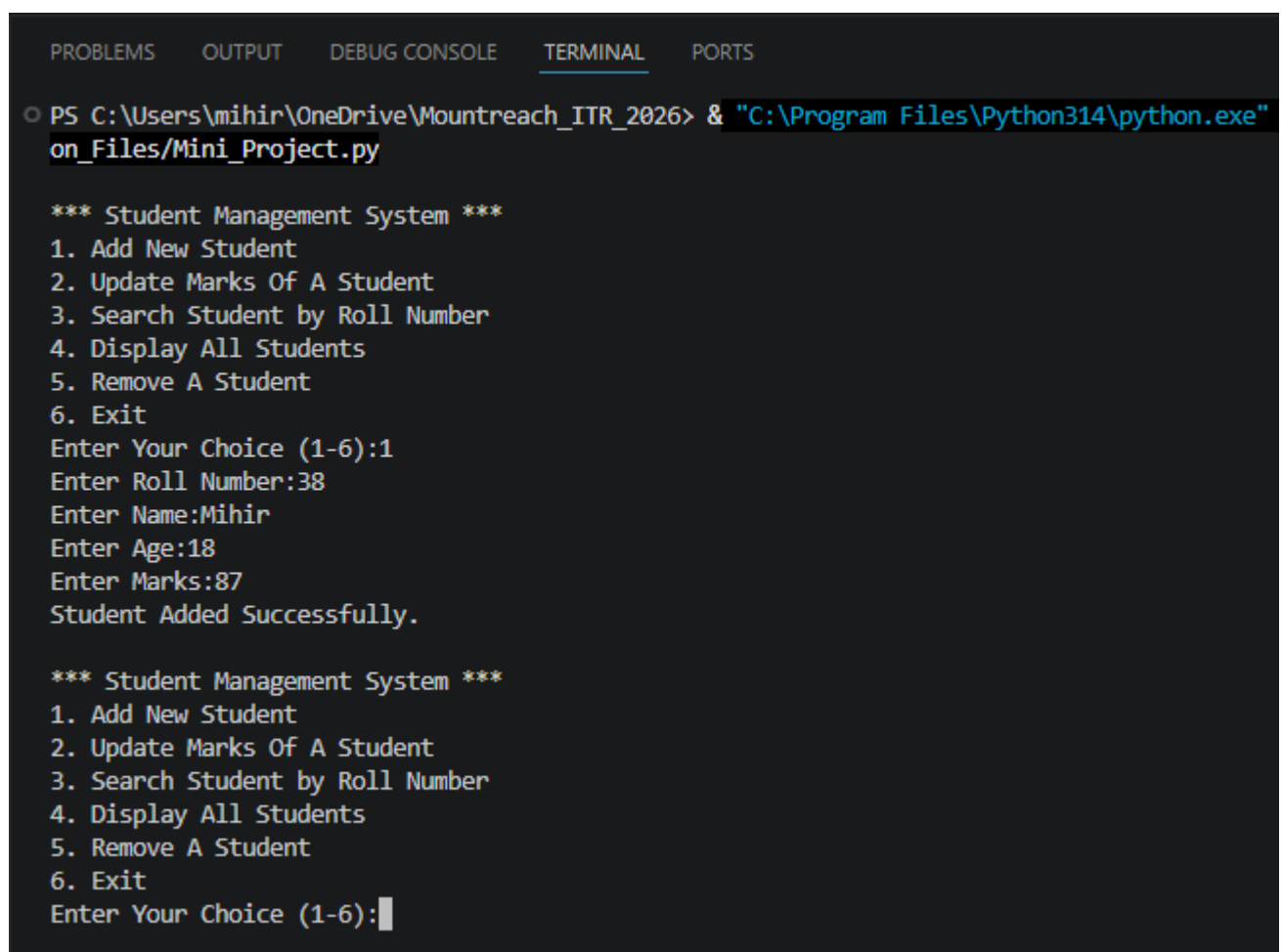
10. Mini Project - Combined Concepts

Create a **Student Management System** using a dictionary:

- Repeatedly ask the user to enter student details (roll number as key, and a nested dictionary containing name, age, and marks).
- Use a while loop with a menu:
 - a) Add new student
 - b) Update marks of a student
 - c) Search student by roll number
 - d) Display all students
 - e) Remove a student
 - f) Exit

Use appropriate dictionary methods (get(), update(), pop(), etc.). Add meaningful comments

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\mihir\OneDrive\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe" on_Files/Mini_Project.py

*** Student Management System ***
1. Add New Student
2. Update Marks Of A Student
3. Search Student by Roll Number
4. Display All Students
5. Remove A Student
6. Exit
Enter Your Choice (1-6):1
Enter Roll Number:38
Enter Name:Mihir
Enter Age:18
Enter Marks:87
Student Added Successfully.

*** Student Management System ***
1. Add New Student
2. Update Marks Of A Student
3. Search Student by Roll Number
4. Display All Students
5. Remove A Student
6. Exit
Enter Your Choice (1-6):
```